

RESEARCH HIGHLIGHTS

- 3D multi-physics model couples 100 GW optics, FEA mechanics, and radiation heat. (Length: 80 chars)
- ASME GCI of 0.13% and $p=1.95$ confirm rigorous numerical grid independence. (Length: 74 chars)
- Membrane stress of 530.88 MPa maintains a 3.77x safety factor in Si₃N₄ substrate. (Length: 81 chars)
- Thermal equilibrium holds core at 923.02 K with a +1247 K margin below sublimation. (Length: 83 chars)
- Photon thrust model matches JAXA IKAROS flight telemetry within 0.12% error. (Length: 76 chars)